Yuyan (Una) Wu

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EDUCATION

Brown University | ScM in Computer Science **Expected May 2027**

- **GPA: 4.0/4.0**
- **Relevant Coursework:** Special Topics in Computational Design and Fabrication, Advanced Computer Graphics

Cornell University | BA in Information Science and Cognitive Science **August 2021 – May 2025**

- **GPA: 3.94/4.0** *with distinction in all subjects*
- **Relevant Coursework:** Human-Computer Interaction Design, Object-Oriented Programming and Data Structure, Intro to Machine Learning, Computer Graphics

PUBLICATION

- **JointStudio: Boundary-Centric Design of Decorative Interlocking Millable Joints.**
  Yuyan Wu, Aditya Ganeshan, Maria Larsson, Ariel Shamir, Daniel Ritchie, Takeo Igarashi.
  (Under review at *ACM CHI 2027*)

RESEARCH EXPERIENCE

Accessible CAD **09/2026 – Present**

Graduate Research Assistant | Computational Design Group **Advisor: Adriana Schulz**

- Investigating nonvisual, feature-based CAD authoring for blind and low-vision users, focusing on the specification and verification of geometric references.
- Exploring natural language and tactile interaction to support face, edge, and vertex selection during incremental modeling.

JointStudio **09/2025 – 09/2026**

Research Intern at University of Tokyo | User Interface Lab **Advisor: Takeo Igarashi**

Graduate Research Assistant | Brown Visual Computing **Advisor: Daniel Ritchie**

- Developed a web-based system for designing decorative interlocking joints using React, TypeScript, Three.js, and Open CASCADE.
- Implemented shared-boundary modeling, controls for cutting direction and depth, and CNC fabrication feedback on corner radii and minimum feature size.
- Conducted a formative study with 5 CAD and fabrication practitioners and a user study with 7 novice joinery designers to inform and evaluate the workflow.
- Fabricated prototypes with a desktop CNC mill to assess manufacturability and assembly.
- Led manuscript preparation for a first-author submission.

Virtual Baby **02/2022 – 05/2024**

Undergraduate RA | Eleanor J. Gibson Laboratory of Developmental Psychology **Advisor: Michael Goldstein**

- Designed an experiment to assess the ecological validity of a VR game for studying behavioral responses in research on parental perception under varying stress levels.
- Developed a participant-training tutorial in Unity3D and C#; modeled, rigged, and animated 3D assets using Unity and Blender.
- Ran the VR study with approximately 50 participants and preprocessed study data using Excel macros and R.

PROJECT EXPERIENCE

Undergraduate Research Assistant | Cornell Program of Computer Graphics **Supervisor: Donald Greenberg**

3D Scanning and House Simulation Website **05/2024 – 07/2024**

- Developed a Three.js website integrating 2D floor plans, 3D room reconstructions, and manipulation of scanned furniture models.
- Tested an initial workflow with Kendal at Ithaca, a retirement community, combining reconstructed 3D rooms and scanned furniture models to help retired faculty decide which furniture to bring an dhow to arrange it in their new homes.

Physics-Based Virtual Grasping **02/2024 – 05/2024**

- Implemented VR hand-object interactions using Unreal Engine, including joint-level collision detection using customized sphere colliders; and using dynamic force calculations based on object velocity to improve grasp realism without haptic feedback.

Design Lead | Game Design Initiative at Cornell **Supervisor: Walker White**

S.A.M.S.A.: desktop escort platformer **02/2024 – 05/2024**

- Created and maintained style guides, visual design specification, and asset management pipelines to streamline production and ensure visual consistency.
- Designed all HUD and UI components and background art in the game.
- Led design critiques and provided conceptual guidance and individual mentorship to the design team.
- Collaborated closely with the programming team to translate initial concept art to final visual implementation; and assisted in implementing the HUD/UI and character animations using LibGDX.

TEACHING EXPERIENCE

Head Design Teaching Assistant **Spring 2025**

CS/INFO 3152 | Introduction to Game Architecture and Design

- Managed game design assignments and tutorials; led design critique and held office hours.

Teaching Assistant **Fall 2024**

CS 1602 | Visual Imaging in Electronic Age

- Assisted in assignment design and grading for the Photogrammetry unit; led weekly recitations and held office hours for approximately 20 students.

Teaching Assistant **Spring 2024**

CS 1340 | Choices and Consequences in Computing

- Provided office-hour support to approximately 150 students on the mathematics and algorithms in weekly assignments; participated in weekly grading sessions.

Teaching Assistant **Fall 2022 and Fall 2023**

INFO 1300 | Introductory Web Design and Programming

- Led weekly discussion sections on user-centered design and front-end web development for approximately 20 students; held office hours and participated in weekly grading sessions.

Teaching Assistant **Spring 2023**

STSCI 2100 | Introductory Statistics

- Led weekly discussions and software demonstrations on fundamental statistical concepts for approximately 15 students; held office hours to support students with course material.

SKILLS

- **Programming:** Python, Java, R, C++, C#, HTML, CSS, JavaScript, TypeScript
- **Development and Design:** React, Three.js, Open CASCADE, Blender, Figma, Unity, Unreal Engine
- **Fabrication:** CNC milling, 3D printing, laser cutting, woodworking
- **CAD/CAM:** Autodesk Fusion, VCarve